Reclassification and Axiomatization of Chromosomal Anomaly Diseases in the Mondo Ontology

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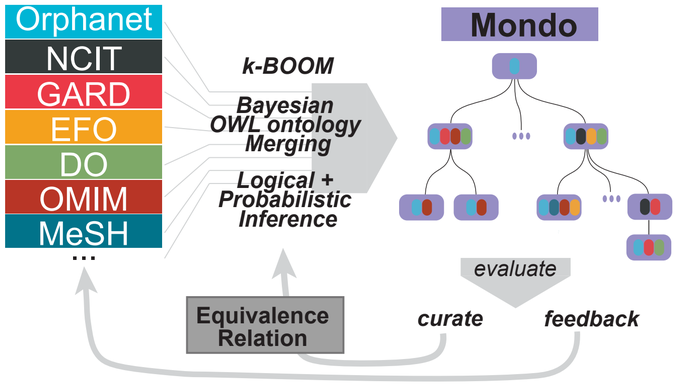
mondo

THE WORLD'S DISEASE CONCEPTS UNIFIED

The multitude of disease ontologies and terminologies poses challenges to data integration; in particular chromosomal anomalies are currently poorly represented in data, posing analytical challenges to research in Down Syndrome and other chromosomal anomalies.
The Mondo Disease Ontology integrates multiple disease tontologies/terminologies into a coherent logic-based ontology that provides precise semantic mappings between terms. It uses Design Patterns to consistently and automatically apply equivalence and subclass axioms to categories of diseases, allowing automatic classification and inferencing.

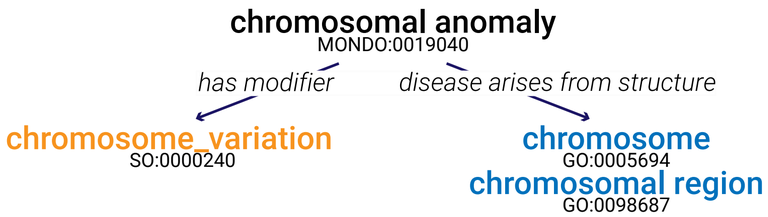


To faithfully represent chromosomal anomalies in Mondo, the axioms make use of Monochrome, an ontology created by automatic translation of UCSC chromosome bands to OWL classes. We recently developed Monochrom and used it to refactor the 'chromosomal anomaly' branch of Mondo.

Chromosomal Anomaly Patterns

Patterns include axioms representing

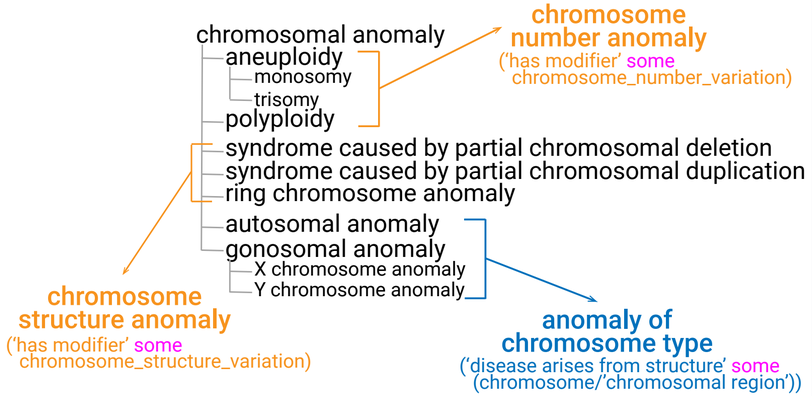
- **type of chromosomal anomaly** (Sequence Ontology term)
- **chromosome/chromosomal region** at the root of the disease (Monochrome term)



Example: 'chromosome 10q23 deletion syndrome'
EquivalentTo:
'chromosomal anomaly' and ('has modifier' some chromosomal_deletion) and ('disease arises from structure' some 'chr10q23 (Human)')

Chromosomal Anomaly Classification

'chromosomal anomaly' terms were classified based on concepts (**bold**) defined by designed patterns.



Disclaimer: for display reasons, not every terms are represented in the classification shown here.

Take Homes

- **Mondo** is an open unified ontology of disease concepts and is available for use in all annotations or for data integration tasks.
- **Monochrom** is ready for use and is pending release; feedback is welcome.
- The use of **design patterns** allowed clear definition of 'chromosomal anomaly' terms and consistent classification of this class.
- This ongoing work will support **accurate cross-referencing** between databases/terminology sources, and **harmonization** of chromosomal anomaly, such as Down Syndrome, data.

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github.com/monarch-initiative/mondo
github.com/monarch-initiative/monochrome